



NAME: _____

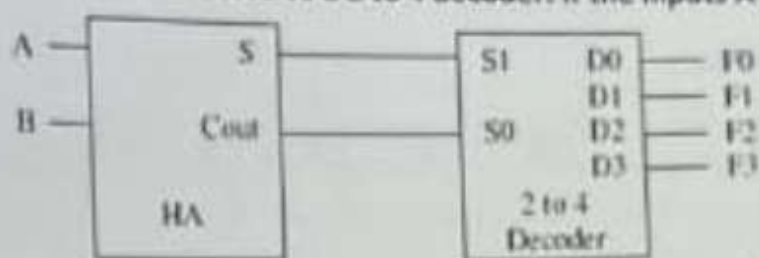
Question 1: Multiple choice Question (30 marks)

1. A combinational circuit that selects one from many inputs are _____
(a) Encoder (b) Decoder (c) Demultiplexer (d) Multiplexer
2. A combinational circuit is one in which the output depends on the _____
(a) Input combination at the time
(b) Input combination and the previous output
3. _____
(a) Input combination at that time and the previous input combination
(b) Input combination and the previous output
(c) Input combination at that time and the previous input combination
(d) Present output and the previous output
4. In a multiplexer the output depends on its _____
(a) Data inputs (b) Select inputs (c) Select outputs (d) Enable pin
5. In 1-to-4 multiplexer, if $C1 = 0$ & $C2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
6. In 1-to-4 multiplexer, if $C1 = 1$ & $C2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
7. Which of the following is not a sequential circuit?
(a) Flip-flop (b) Counter (c) Shift Register (d) Multiplexer
8. The sequential circuit the output state depend upon _____
(a) Past input state (b) Present input state (c) Present as well as past input (d) None of the above
9. A 4 bit counter will count _____
(a) 0 - 3 (b) 0 - 4 (c) 0 - 15 (d) 0 - 16
10. The following truth table represents a _____ gate

X	Y	F
0	0	1
0	1	0
1	0	0
1	1	1

 (a) NOR (b) HALF ADDER (c) XNOR (d) XOR

9) In the circuit below, a Half-Adder is connected to a 2 to 4 decoder. If the inputs $A=1$ and $B=1$, then:



- a) $F_0=1$ b) $F_1=1$ c) $F_2=1$ d) $F_3=1$

10) The minimum sum of product expression for F from the K-map given below

$AB \backslash CD$	00	01	11	10
00	1	X		1
01	1	1	1	1
11		1	1	X
10			X	X

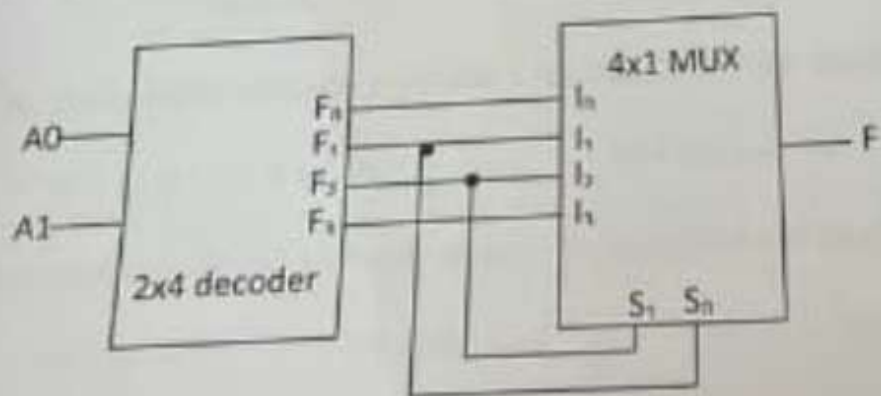
- a) $A'D' + BD + CD'$
 b) $A'D' + BD$
 c) $A'C' + BD + CD'$
 d) $BD + CD'$

11) Simplification using the k-map

$AB \backslash CD$	00	01	11	10
00	1	1	1	1
01	1	1	1	
11				
10	1			1

- a) $a'b' + a'c' + da' + b'd'$
 b) $a'b' + da' + d'b'a$
 c) $a'b' + d'b'a + a'c' + da'$
 d) $a'c' + da' + b'd'$

12) The 2x4 decoder has inputs A_1 and A_0 . what is the output of multiplexer F if $A_1=1$ and $A_0=1$



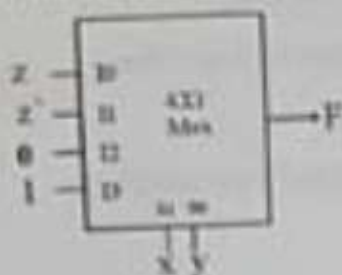
- a) $F=0$
 b) $F=1$

11. The combinational circuit represents



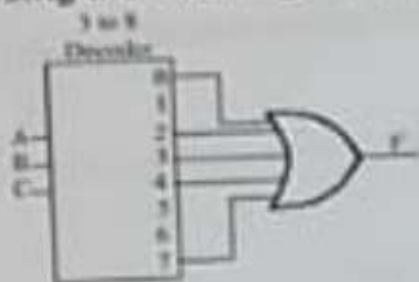
- (a) Decoder
- (b) Encoder
- (c) Comparator
- (d) Multiplexer

12. The function represented by the following 4X1 Multiplexer is



- (a) $F = x'yz + x'y'z' + xy$
- (b) $F = xy + xz + x'y'z$
- (c) $F = x'y'z + x'yz' + xy$
- (d) $F = x'y'z + xy'z' + x'y$

13. Using the following Decoder, the simplified SOP for the function F is:



- (a) $F = B'C' + BC + AB'$
- (b) $F = B'C + BC + AB'$
- (c) $F = BC' + BC + AB'$
- (d) $F = B'C' + BC + A'B$

14. To implement $A \oplus B$ using XOR with three inputs, the third input should be

- a) Ground
- b) To one of inputs (A,B)
- c) +5V
- d) Answers b or c

15. Which one of these called universal gate:

- a) NAND only
- b) NAND, NOR
- c) AND, OR, NOT
- d) OR, AND, NOT, NOR, XOR, XNOR, NAND

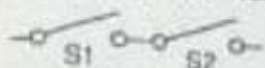
16. What is the output of XOR with two inputs A,B, if A is connected to +5v:

- a) Always zero
- b) Always one
- c) B
- d) B'

Q1) multiple choice :

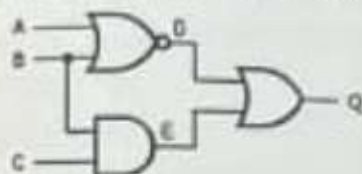
1	2	3	4	5	6	7	8	9	10	12	13	14	15

1) The logical gate that gives you the function of two switches in series is:



- a) OR Gate b) AND Gate c) XOR Gate d) None of them

2) The following logic circuit gives the function $F = \dots\dots\dots$



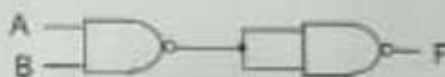
- a) $(A+B) + B.C$ b) $(A+B)' + (B'+C)'$ c) $(A'+B).(B+C)'$ d) None of them

3) The correct leg numbering for the following IC is $\dots\dots\dots$ starting from the mark on it.



- a) Clockwise b) Counter Clockwise c) 1 and 2 are the same d) None of them

4) which gate the following circuit can represent



- a) AND b) OR c) XOR d) NOR

5) Simplification of the Boolean expression $AB + ABC + ABCD + ABCDE + ABCDEF$ yields which of the following results?

- (a) ABCDEF (b) AB (c) $AB + CD + EF$ (d) $A + B + C + D + E + F$

6) Given that $F = (A + B' + C)(D + E)$, which of the following represents the only correct expression for F' ?

- (a) $F' = A'BC' + D' + E'$ (b) $F' = AB'C + DE$ (c) $F' = (A' + B + C')(D' + E')$ (d) $F' = A'BC' + D'E'$

7) Strobe S in a multiplexer acts as

- A) Reset B) Enable C) Clear D) Stop

8) Which of the following Boolean expressions are false?

- a) $(X+Y)(X+Z)=X+YZ$ b) $(X+Y)(X'+Z)=X'Y+XZ$ c) $XY+X'=X'+Y$ d) $X+(Y+Z)=(X+Y)+Z$ e) none of the above